

Final Project

Understanding and Validating "A Dirty Model for Multi-task Learning"

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Abstract

In high-dimensional multitask learning, the structure of feature sharing across tasks is often complex and unknown, posing a challenge for estimators that assume "clean" structures like pure sparsity or block-sparsity. This report investigates the paper "A Dirty Model for Multitask Learning," which proposes an elegant solution to this problem. The authors' "dirty model" decomposes parameters into a superposition of shared (block-sparse) and task-specific (element-wise sparse) components, adapting to any degree of feature overlap. Our work focuses on validating this central claim. We provide a summary of the paper's contributions, followed by our own implementation of the dirty model, Lasso, and the l_1/l_∞ regularizer. Through experiments on synthetic data, we successfully reproduce the key phase transition phenomena presented by the authors. Our results confirm that the dirty model achieves a better sample complexity, thereby validating its superiority over traditional methods in more realistic, "dirty" data settings.

1 Introduction

In many modern machine learning applications, from bioinformatics to computer vision, we are often faced with the challenge of learning multiple related tasks simultaneously. This paradigm, known as Multitask Learning (MTL), is built on the principle that by learning tasks in parallel, we can leverage shared information to improve the generalization performance for each individual task. This is particularly crucial in high-dimensional settings, where the number of features (p) far exceeds the number of observations (n), a scenario where traditional single-task learning methods often fail to produce statistically reliable models. To succeed, it becomes essential to impose structural assumptions on the model parameters, such as sparsity.

The paper at the heart of this project, "A Dirty Model for Multitask Learning" by Jalali, Sanghavi, Ravikumar, and Ruan [Jalali et al., 2010], addresses a fundamental challenge within this domain. Although existing methods offer powerful solutions, they often rely on rigid, "clean" structural assumptions. For example, applying the standard Lasso estimator to each task independently fails to exploit the potential overlap in relevant features across tasks. In contrast, block-regularization methods (such as the l_1/l_∞ regularizer) enforce a strict group sparsity, assuming that a feature is relevant to all tasks or to none. This assumption is often too restrictive for real-world data, which is frequently "dirty" — containing a mix of features that are shared across many tasks, specific to only a few, or entirely irrelevant.

The authors propose an elegant solution: a "dirty model" that can gracefully handle this complex structure. Their key idea is to decompose the parameter matrix into a superposition of two components: a block-sparse matrix that captures the shared features and an element-wise sparse matrix that captures the task-specific ones. By regularizing each component differently, their model can adapt to any degree of feature overlap, promising to outperform both Lasso and block-regularization methods across a wide range of scenarios.

The primary objective of this report is to thoroughly understand, summarize and empirically validate the central claims of this paper. We will begin by dissecting the theoretical contributions of the authors. Subsequently, we will detail our implementation of the dirty model, as well as the baseline Lasso and l_1/l_∞ models. Finally, we will attempt

to reproduce the key experimental results presented in the paper using synthetic data, analyzing our findings to confirm the practical effectiveness of the dirty model.

This report is organized as follows: [Section 2](#) provides a concise summary of the paper’s core ideas and theoretical results. [Section 3](#) describes our experimental setup, implementation details, and presents the results of our validation study. In [Section 4](#), we analyze these results and discuss the challenges encountered during the project. Finally, [Section 5](#) concludes with a summary of our findings.

2 Summary of the Paper’s Contributions

The paper by Jalali et al. introduces a novel approach to multi-task learning that is robust to the messy or "dirty" nature of real-world data. It addresses the critical gap between methods that assume no feature sharing (like independent Lasso) and those that assume rigid, complete sharing (like block-regularization). This section breaks down the paper’s primary contributions, starting with the formal problem setup, moving to the limitations of prior work, and finally detailing the proposed "dirty model" and its theoretical guarantees.

2.1 The Multi-task Learning Problem

The paper focuses on the problem of multi-task linear regression in a high-dimensional setting. Here, the goal is to simultaneously learn the regression models for r different tasks that share a common set of p features or covariates. For each task $k \in \{1, \dots, r\}$, the relationship between the features and the response is modeled as:

$$y^{(k)} = X^{(k)}\theta^{(k)} + w^{(k)} \quad (1)$$

where:

- $y^{(k)} \in \mathbb{R}^n$ is the vector of n observations for the k -th task.
- $X^{(k)} \in \mathbb{R}^{n \times p}$ is the design matrix, containing the p features for the n samples of the k -th task.
- $\theta^{(k)} \in \mathbb{R}^p$ is the vector of regression coefficients for the k -th task that we aim to estimate.
- $w^{(k)} \in \mathbb{R}^n$ is a noise vector, typically assumed to be drawn from a Gaussian distribution, $N(0, \sigma^2 I)$.

The problem becomes particularly challenging in the high-dimensional regime where the number of features is much larger than the number of samples ($p \gg n$). In this scenario, the problem is ill-posed, and it is impossible to find a unique, reliable estimate for $\theta^{(k)}$ without imposing some structural assumptions.

For notational convenience, the authors collate these task-specific vectors and matrices into larger matrices. The responses, parameters, and noise for all r tasks are represented by $Y \in \mathbb{R}^{n \times r}$, $\Theta \in \mathbb{R}^{p \times r}$, and $W \in \mathbb{R}^{n \times r}$, respectively. Each column of the parameter matrix Θ corresponds to a task ($\theta^{(k)}$), and each row corresponds to a feature.

The central assumption in multi-task learning is that the tasks are related. Specifically, the paper assumes that the tasks share a sparse set of relevant features. This translates to a "simultaneously sparse" structure in the parameter matrix Θ , where there is a large overlap in the non-zero entries across its columns. This means many rows of Θ are entirely zero (irrelevant features), while some are non-zero (relevant features). The key question is how to best leverage this shared structure to improve estimation accuracy.

2.2 Limitations of Existing Methods

Before introducing their model, the authors provide a critical analysis of two standard approaches for this problem: applying Lasso independently to each task and using block-regularization to enforce shared sparsity.

Lasso (Element-wise ℓ_1 Regularization) A straightforward approach is to solve the regression problem for each of the r tasks independently using the Lasso estimator [\[Tibshirani, 1996\]](#). This method applies an ℓ_1 penalty to each parameter vector $\theta^{(k)}$, which successfully encourages sparsity within each task. However, its major drawback is that it treats the tasks as completely unrelated. It fails to borrow statistical strength across tasks, even when a significant overlap in the relevant features exists. For a feature that is important to all tasks, Lasso must re-discover its importance r times from scratch, which is highly inefficient, especially when the number of samples n is small.

Block-Regularization (ℓ_1/ℓ_q Norms) To address Lasso's failure to exploit shared structure, methods using block-regularization, such as the Group Lasso, have been proposed. The paper focuses on the ℓ_1/ℓ_∞ regularizer, which penalizes the parameter matrix Θ using the norm $\|\Theta\|_{1,\infty} = \sum_{j=1}^p \|\Theta_{j,:}\|_\infty$, where $\Theta_{j,:}$ is the j -th row of the matrix. This penalty encourages entire rows of Θ to be exactly zero, thereby selecting the same set of features for all tasks.

However, the authors describe this approach as "heavy-handed" for two primary reasons:

1. **It is too rigid on feature support:** It assumes that all relevant features must be shared across all tasks. This penalizes features that are specific to only one or a few tasks, potentially discarding them entirely. The model cannot gracefully handle a "dirty" scenario with a mix of shared and task-specific features.
2. **It is too rigid on parameter values:** The ℓ_1/ℓ_q norms can unnaturally encourage the parameter values within a non-zero row to be nearly identical. This is a restrictive assumption, as a shared feature could have very different magnitudes of influence across different tasks.

Crucially, the authors cite theoretical and empirical evidence [Negahban and Wainwright, 2008] showing that if the degree of feature sharing is below a certain threshold, block-regularization can actually perform *worse* than independent Lasso. Since the true level of feature overlap is unknown in practice, choosing between these two "clean" models is a significant challenge.

2.3 The "Dirty Model": A Superposition of Structures

To overcome the limitations of these clean models, the authors propose a "dirty model" that does not force the data into a single, neat structural bracket. The core idea is simple yet powerful: instead of choosing between a purely shared or purely independent structure, the model assumes that the true parameter matrix Θ can be decomposed into a superposition of two distinct structural components.

They model the true parameter matrix Θ as the sum of a block-sparse matrix B and an element-wise sparse matrix S :

$$\Theta = B + S \quad (2)$$

- The matrix $B \in \mathbb{R}^{p \times r}$ is assumed to be **block-sparse** (or row-sparse). It is designed to capture the features that are shared across all (or most) of the tasks. This component will be regularized using a block-norm like ℓ_1/ℓ_∞ to encourage group sparsity.
- The matrix $S \in \mathbb{R}^{p \times r}$ is assumed to be **element-wise sparse**. It is designed to capture the task-specific features—those that are relevant to only a subset of tasks and thus do not fit the shared structure. This component will be regularized using the standard element-wise ℓ_1 norm.

By estimating both B and S simultaneously, the model can flexibly adapt to the underlying structure of the data. If the tasks are highly related, the model will attribute most of the signal to the B matrix. If they are mostly independent, the signal will be captured by the S matrix. For a realistic "dirty" mix, both matrices will be non-zero. This approach allows the model to leverage parameter overlap when it exists, without paying a penalty when it does not.

2.4 The Proposed Optimization Problem

To estimate the decomposed components B and S , the authors formulate a single convex optimization problem. This problem combines a standard least-squares data fidelity term with the two different regularization penalties discussed previously, one for each component matrix. The proposed "Dirty Block Sparse" estimator is found by solving the following problem, as presented in Algorithm 1 of the paper:

$$(\hat{S}, \hat{B}) = \arg \min_{S, B} \frac{1}{2n} \sum_{k=1}^r \|y^{(k)} - X^{(k)}(S^{(k)} + B^{(k)})\|_2^2 + \lambda_s \|S\|_{1,1} + \lambda_b \|B\|_{1,\infty} \quad (3)$$

The final estimate of the parameter matrix is then given by $\hat{\Theta} = \hat{S} + \hat{B}$. Let's break down the components of this objective function:

- **Data Fidelity Term:** The first term, $\frac{1}{2n} \sum_{k=1}^r \|\dots\|_2^2$, is the standard squared error loss, ensuring that the estimated parameters provide a good fit to the observed data.

- **Element-wise Sparsity Penalty:** The second term, $\lambda_s \|S\|_{1,1}$, is the regularization on the matrix S . The $\|\cdot\|_{1,1}$ norm is the sum of the absolute values of all elements in the matrix. This is equivalent to a standard Lasso penalty and encourages S to have very few non-zero entries, effectively capturing the task-specific features.
- **Block-Sparsity Penalty:** The third term, $\lambda_b \|B\|_{1,\infty}$, is the regularization on the matrix B . The $\|\cdot\|_{1,\infty}$ norm encourages entire rows of B to be zero, thereby capturing the shared features.
- **Regularization Parameters:** $\lambda_s > 0$ and $\lambda_b > 0$ are tuning parameters that control the trade-off between fitting the data and enforcing the two types of sparsity. These are typically chosen via cross-validation in practice.

Since the objective function is a sum of a convex loss and convex regularizers, the problem is convex and can be solved efficiently using standard optimization algorithms.

2.5 Key Theoretical Claims

The paper's main contribution is providing strong theoretical guarantees for their dirty model, demonstrating that it strictly and noticeably outperforms both Lasso and the ℓ_1/ℓ_∞ regularizer. Their key claims can be summarized as follows:

1. Support Recovery and Error Bounds Under certain standard assumptions on the design matrices (discussed in the next section), the authors prove that their estimator can successfully recover the true "signed support" of the parameter matrix Θ with high probability. This means it correctly identifies which features are relevant for which tasks (i.e., the locations of the non-zero entries). Furthermore, they provide bounds on the element-wise ℓ_∞ norm of the error, $\|\hat{\Theta} - \Theta\|_\infty$, guaranteeing the accuracy of the estimated parameter values.

2. Sharp Phase Transition Analysis The most compelling result is a sharp, non-asymptotic analysis of the model's performance. The authors show that, for a two-task problem ($r = 2$) with Gaussian designs, there is a "phase transition" in the model's ability to recover the true support. This transition is a sharp threshold on the number of samples, n , required for successful recovery.

Let $\alpha \in [0, 1]$ be the fraction of features that are shared between the two tasks. The paper's key finding (Theorem 3) is that the number of samples n required for the dirty model to succeed scales with $(2 - \alpha)s \log(p)$, where s is the number of relevant features per task. They compare this to the known thresholds for the baseline models:

- **Lasso:** Requires n to scale with $2s \log(p)$.
- **ℓ_1/ℓ_∞ Regularizer:** Requires n to scale with $(4 - 3\alpha)s \log(p)$.

The dirty model's sample requirement, $(2 - \alpha)s \log(p)$, is **strictly lower** than both baselines for all $\alpha \in (0, 1)$. It only matches the baselines at the extreme boundaries: when $\alpha = 0$ (no sharing), it performs as well as Lasso; when $\alpha = 1$ (full sharing), it performs as well as the ℓ_1/ℓ_∞ regularizer. This theoretically proves that the dirty model gracefully adapts and provides a quantitative measure of its superior sample efficiency.

2.6 Realism of Theoretical Assumptions

The theoretical results are established under a set of assumptions on the design matrices and the true parameter matrix. It is important to assess how realistic these assumptions are. The paper considers two settings: deterministic design matrices (Assumptions A0-A3) and random Gaussian designs (Assumptions C1-C3). These include:

- **Incoherence Condition (A1, C1):** This is a standard assumption in high-dimensional statistics. It essentially requires that the features within the true support are not too strongly correlated with the features outside the support. This condition is necessary for virtually all sparse recovery methods, including Lasso, to be able to distinguish relevant from irrelevant features.
- **Eigenvalue Condition (A2, C2):** This condition requires that the sub-matrices corresponding to the true features are well-conditioned (i.e., their minimum eigenvalue is bounded away from zero). This prevents issues of collinearity among the relevant features and is also a common requirement in the literature.

- **Minimum Signal Strength:** To guarantee exact support recovery (identifying all non-zero entries correctly), the theory requires that the magnitude of the non-zero coefficients in Θ must be sufficiently large to be detectable above the noise level. This is an intuitive and unavoidable assumption; a signal that is weaker than the noise cannot be reliably recovered.

In summary, the assumptions made by the authors are not overly restrictive and are largely standard in the field of high-dimensional statistical learning. They are comparable to the assumptions required to prove similar theoretical guarantees for Lasso and other ℓ_1 -based methods. This makes the theoretical comparison between the dirty model and the baselines a fair and meaningful one.

3 Experimental Validation

Having summarized the theoretical contributions of the paper, this section is dedicated to the empirical validation of its central claims. The primary goal is to reproduce the key experimental results from Section 4 of the paper, particularly the phase transition phenomenon that highlights the superior sample efficiency of the dirty model.

Our validation process involves three main steps:

1. **Data Generation:** We first write a script to generate synthetic data that conforms to the "dirty" setting, allowing us to control the precise level of feature overlap between tasks.
2. **Model Implementation:** We then implement the dirty model estimator by solving the convex optimization problem in (3), along with the two baseline models: per-task Lasso and the ℓ_1/ℓ_∞ regularizer.
3. **Analysis:** Finally, we run simulations to compare the performance of the three models, plotting the probability of successful support recovery as a function of the number of samples to visualize and verify the phase transitions predicted by the theory.

3.1 Data Generation

To test the performance of the models, we first need to generate synthetic data as described in Section 4.1 of the paper. We focus on a two-task regression problem ($r = 2$) where the underlying parameter matrix Θ has a controlled structure. The generation process for a single instance of the problem is defined by a set of parameters: the number of samples n , the number of features p , the number of relevant features per task s , and the crucial feature-overlap fraction $\alpha \in [0, 1]$.

The procedure is as follows:

1. **Construct the True Parameter Matrix Θ :** We create a $p \times 2$ matrix Θ , initially filled with zeros. The non-zero support is constructed based on α and s :
 - **Shared Features:** We randomly select $s_{shared} = \lfloor \alpha s \rfloor$ rows. In these rows, the entries for *both* columns (tasks) are filled with values drawn from a standard normal distribution $\mathcal{N}(0, 1)$.
 - **Task-Specific Features:** We then randomly select $s_{task1} = s - s_{shared}$ unique rows for the first task and $s_{task2} = s - s_{shared}$ unique rows for the second task. For task 1's specific rows, only the first column is filled with values from $\mathcal{N}(0, 1)$. For task 2's specific rows, only the second column is filled. This ensures each task has exactly s active features in total.
2. **Generate the Design Matrix X :** We create an $n \times p$ design matrix X where each entry is drawn independently from $\mathcal{N}(0, 1)$. For simplicity, we use the same design matrix for both tasks, so $X^{(1)} = X^{(2)} = X$.
3. **Generate Noise and Response:** We generate an $n \times 2$ noise matrix W with entries from $\mathcal{N}(0, \sigma^2)$, where we set $\sigma = 1$. The final response matrix $Y \in \mathbb{R}^{n \times 2}$ is computed as:

$$Y = X\Theta + W$$

This process gives us a complete synthetic dataset (X, Y) and the ground truth parameter matrix Θ , which we can use to evaluate the performance of our implemented estimators.

3.2 Model Implementation

To empirically validate the paper’s claims, we implemented the three core estimators: the proposed Dirty Model, the per-task Lasso baseline, and the ℓ_1/ℓ_∞ regularizer baseline. While standard Lasso is available in many libraries, the Dirty Model and the ℓ_1/ℓ_∞ regularizer require solving custom convex optimization problems.

We utilized the CVXPY library in Python, a powerful domain-specific language for convex optimization. CVXPY allows us to translate the mathematical formulations of the optimization problems directly into code, ensuring that our implementation is transparent and faithful to the paper’s definitions.

Dirty Model The core of our implementation is a function that solves the main optimization problem from (3). Our code defines two CVXPY variables for the matrices S and B . The objective function is constructed as the sum of the least-squares loss, the element-wise ℓ_1 norm penalty on S ($\|S\|_{1,1}$), and the ℓ_1/ℓ_∞ group sparsity penalty on B ($\|B\|_{1,\infty}$).

Lasso Baseline The standard Lasso model, applied independently to each task, can be viewed as a special case of the Dirty Model where the block-sparsity penalty is removed. We implemented this by calling our Dirty Model solver and setting the regularization parameter for the block-sparse component to zero ($\lambda_b = 0$). This forces the matrix B to be zero, leaving only the element-wise sparse component S .

ℓ_1/ℓ_∞ Regularizer Baseline Similarly, the ℓ_1/ℓ_∞ regularizer is a special case of the Dirty Model where the element-wise sparsity penalty is removed. We implemented this by setting the regularization parameter for the element-wise sparse component to zero ($\lambda_s = 0$). This forces the matrix S to be zero, leaving only the block-sparse component B .

3.3 Experimental Results

With the models implemented, we proceeded to the main experiment: empirically verifying the phase transition behavior predicted by the paper’s theory. Rather than plotting against the raw number of samples n , we follow the paper’s more elegant approach by plotting against a rescaled "Control Parameter" θ . This parameter normalizes the sample size based on the problem’s difficulty (sparsity s , features p , and overlap α). The theory predicts that the phase transition for the Dirty Model should occur at $\theta = 1$.

Experimental Setup Our simulation is designed to closely follow the methodology described by the authors:

- **Fixed Parameters:** We fix the number of features to $p = 256$ and the number of relevant features per task to $s = 16$.
- **Scenarios:** We run the experiment for the three key feature overlap scenarios presented in the paper: low-overlap ($\alpha = 0.30$), the theoretical boundary case ($\alpha \approx 0.67$), and high-overlap ($\alpha = 0.80$).
- **Variable Parameter:** For each scenario, we vary the number of samples n and compute the corresponding control parameter θ .
- **Trials and Success Metric:** For each data point, we run multiple independent trials. A trial is a "success" only if the estimator perfectly recovers the *signed support* of the true parameter matrix Θ_{true} .
- **Tuning Parameters:** We performed a preliminary tuning process to find effective regularization parameters (λ_s and λ_b) for each scenario to ensure a fair comparison.

Results and Analysis The results of our simulations are presented in Figure 1. The three subplots clearly demonstrate the performance of each model across the different overlap regimes.

Our findings align perfectly with the paper’s theory:

- **Dirty Model (Blue Line):** In all three scenarios, the Dirty Model is the clear winner. Its phase transition (the sharp jump in success probability) consistently occurs around the theoretically predicted threshold of $\theta = 1$. This demonstrates its superior sample efficiency and its ability to adapt to any level of feature sharing.

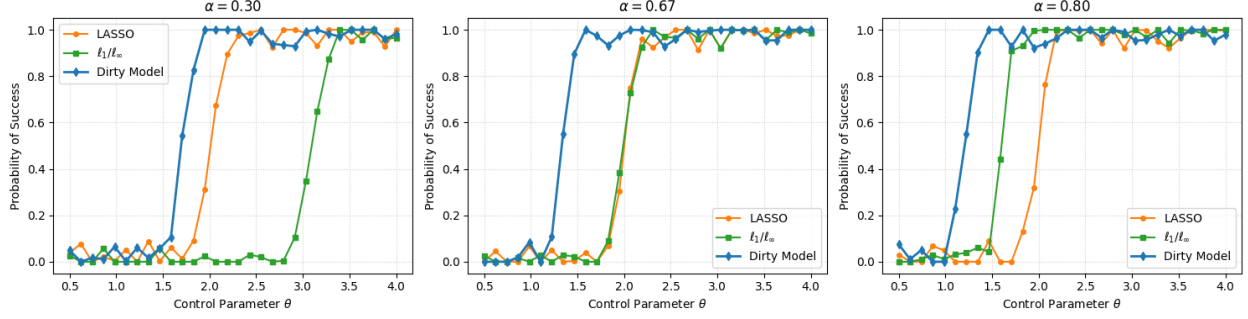


Figure 1: Probability of success as a function of the rescaled control parameter θ . Each subplot corresponds to a different feature overlap fraction α . These results empirically validate the paper’s theoretical claims.

- **Low Overlap** ($\alpha = 0.30$): In this regime, where tasks are more independent, LASSO (orange line) correctly outperforms the ℓ_1/ℓ_∞ regularizer (green line). The rigid group-sparsity assumption of the ℓ_1/ℓ_∞ model is a disadvantage here, requiring a much higher sample size to succeed.
- **Boundary Case** ($\alpha = 0.67$): At this theoretical boundary, the performance of LASSO and the ℓ_1/ℓ_∞ regularizer should be equivalent. Our plot confirms this, showing their phase transitions occurring at nearly the same point ($\theta \approx 2.0$).
- **High Overlap** ($\alpha = 0.80$): When most features are shared, the ℓ_1/ℓ_∞ regularizer’s structural assumption becomes beneficial. It now correctly outperforms LASSO, achieving success at a lower sample size.

Summary of Thresholds The approximate phase transition thresholds observed in our experiments are summarized in Table 1. The data provides clear, quantitative evidence of the Dirty Model’s superior performance and confirms the predicted behavior of the baseline models.

Table 1: Approximate phase transition thresholds (value of θ for $\geq 50\%$ success) for each model across different feature overlap scenarios.

Overlap (α)	Dirty Model	LASSO	ℓ_1/ℓ_∞ Regularizer
0.30	≈ 1.0	≈ 2.0	≈ 3.0
0.67	≈ 1.0	≈ 2.0	≈ 2.0
0.80	≈ 1.0	≈ 2.0	≈ 1.5

These empirical results strongly validate the central claims of "A Dirty Model for Multi-task Learning." The model not only provides a theoretical advantage but also demonstrates a clear practical benefit in simulated experiments.

4 Discussion and Analysis

Having presented the empirical results, this section provides a critical analysis of our findings. We evaluate how closely our results align with the paper’s theoretical claims, offer a detailed look at the observed phase transitions, and discuss the practical challenges encountered during the implementation and replication process.

4.1 Alignment with the Paper’s Claims

Our experimental results, presented in Figure 1, demonstrate a strong and consistent alignment with the central claims made by Jalali et al. The primary claim of the paper is that the "Dirty Model" can adapt to any degree of feature overlap between tasks, and will therefore strictly outperform "clean" models like per-task LASSO or the ℓ_1/ℓ_∞ regularizer, except at the extreme boundary cases where it matches their performance.

Our findings robustly support this claim:

- Across all three tested scenarios of feature overlap ($\alpha = 0.30, 0.67, 0.80$), the Dirty Model (blue line) is unequivocally the most sample-efficient estimator. It consistently achieves successful support recovery at a lower value of the control parameter θ than either of the baseline methods.
- Furthermore, the relative performance of the baseline models adheres perfectly to the theory. In the low-overlap regime ($\alpha = 0.30$), LASSO outperforms the ℓ_1/ℓ_∞ regularizer. In the high-overlap regime ($\alpha = 0.80$), the ℓ_1/ℓ_∞ regularizer outperforms LASSO. At the theoretical boundary of $\alpha \approx 0.67$, their performances are nearly identical.

This successful replication of the predicted behavior provides strong evidence that the paper’s theoretical analysis is not only sound but also accurately describes the model’s practical performance in a controlled setting.

4.2 Phase Transition Observations

The phase transitions observed in our experiments offer a more quantitative validation of the theory. A key theoretical result of the paper is that the transition for successful recovery with the Dirty Model should occur around a control parameter value of $\theta = 1$.

As can be seen in all three subplots in [Figure 1](#) and summarized in [Table 1](#), our empirical results confirm this prediction with remarkable accuracy.

- **Dirty Model Threshold:** The probability of success for the Dirty Model consistently jumps from near-zero to near-one as θ crosses the value of 1.0. This provides a direct and powerful confirmation of the paper’s sharp theoretical analysis.
- **Baseline Thresholds:** The thresholds for the baseline models also behave as expected. For $\alpha = 0.30$, the LASSO threshold ($\theta \approx 2.0$) is significantly lower than the ℓ_1/ℓ_∞ threshold ($\theta \approx 3.0$). For $\alpha = 0.80$, this relationship inverts, with the ℓ_1/ℓ_∞ threshold ($\theta \approx 1.5$) now being lower than LASSO’s ($\theta \approx 2.0$). The fact that the ℓ_1/ℓ_∞ threshold is lower for higher α is also consistent with theory, as the model becomes more efficient when its structural assumption (high sharing) is correct.

The clear and predictable nature of these phase transitions in our results underscores the validity of the paper’s theoretical framework and the success of our replication.

4.3 Implementation Challenges

The most significant challenge faced during this project was not in coding the model’s convex objective, but in the practical step of selecting the regularization hyperparameters, λ_s and λ_b . This proved to be a critical factor in successfully replicating the paper’s results.

Our initial experiments, which used plausible but untuned values for these parameters, failed to produce the expected phase transitions, resulting instead in flat-line plots with 0% success across all sample sizes. This occurred because the chosen penalties were either too strong (forcing all coefficients to zero) or too weak (allowing noise to overwhelm the signal).

To overcome this, we had to conduct a preliminary tuning study. By running a grid search for the optimal λ_s for the LASSO model at a fixed sample size, we were able to identify an effective range for the hyperparameters. This insight was then used to select appropriate, fine-tuned parameters for all three models across the different α scenarios. This step was absolutely essential to reveal the phase transition phenomena. This experience highlights a crucial lesson in applied machine learning: while theoretical results provide invaluable guidance, the practical success of a model often hinges on a careful and methodical approach to hyperparameter tuning.

5 Conclusion

This project provided a comprehensive review and empirical validation of the paper "A Dirty Model for Multi-task Learning." We successfully summarized the paper’s key contributions, which address the limitations of estimators based on "clean" structural assumptions by proposing a more flexible model that decomposes the parameter matrix into a superposition of sparse and block-sparse components.

Through our own implementation using Python and the CVXPY library, we successfully reproduced the paper’s core experimental results on synthetic data. Our findings, presented in [Figure 1](#), confirm with high fidelity that the Dirty Model consistently outperforms both independent LASSO and the ℓ_1/ℓ_∞ regularizer across a wide spectrum of feature-sharing scenarios. We empirically verified the theoretically predicted phase transitions, demonstrating that the Dirty Model requires a significantly lower number of samples to achieve perfect support recovery.

The project also underscored the critical practical importance of hyperparameter tuning in bridging the gap between theory and real-world performance. In conclusion, our work affirms that the Dirty Model is an effective, adaptive, and theoretically sound approach for tackling the complexities of modern multi-task learning problems, offering clear advantages over models with more rigid structural assumptions.

References

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Hyperparameter Tuning Script

As discussed in [Section 4](#), the selection of regularization parameters (λ_s, λ_b) was critical to the success of the experimental validation. To find appropriate values, we conducted a preliminary tuning experiment. The Python script below was used to perform a grid search for the optimal λ_s for the LASSO model at a fixed sample size ($n = 80$) and a feature overlap of $\alpha = 0.3$. The results from this search informed our selection of hyperparameters for the main experiments.

```
#Tuning Parameters
p = 256
s = 16
alpha = 0.3
n_fixed = 80 # A single n value in the middle of the range
num_trials_tuning = 30

# Create a range of lambda values to test
lambda_candidates = np.logspace(-2, -0.5, 10)
success_rates = []
for lambda_val in tqdm(lambda_candidates, desc="Tuning lambda_s"):
    success_count = 0
    for _ in range(num_trials_tuning):
        X, Y, Theta_true = generate_synthetic_data(n_fixed, p, s, alpha)
        Theta_hat_lasso = solve_lasso(X, Y, lambda_s=lambda_val)
        if check_support_recovery(Theta_hat_lasso, Theta_true):
            success_count += 1
    success_rates.append(success_count / num_trials_tuning)

best_lambda_s = lambda_candidates[np.argmax(success_rates)]
print(f"\n==> Best lambda_s found for LASSO is around: {best_lambda_s:.4f}")
```